

You are helping me build a rigorous set of Microsoft Excel exhibits analyzing **PMC methodology activity for life-science tools companies**. Work only on PMC for now. Do not combine PMC with PubMed, bioRxiv, patents, NIH S10, or any composite innovation index. We will conduct cross-source analysis later.

I have already completed Exhibit 1: a 2025 revenue-versus-PMC scatterplot using a log-log regression. Continue from Exhibit 3 through the end of the PMC analysis, including historical data from 2022–2025 and a separate R&D-spend sensitivity analysis.

## Current 2025 worksheet structure

The company data occupy rows 32–54.

| Column | Field                                                 |
|--------|-------------------------------------------------------|
| A      | Ticker                                                |
| B      | Actual 2025 PMC company-publication relationships     |
| C      | 2025 consolidated revenue (\$M)                       |
| D      | LN Revenue                                            |
| E      | LN(1 + Actual PMC)                                    |
| F      | Predicted LN(1 + PMC)                                 |
| G      | Revenue-expected PMC                                  |
| H      | Log residual                                          |
| I      | Actual/expected activity multiple                     |
| J      | Percentage above or below expected                    |
| K      | Actual PMC minus expected PMC                         |
| L      | Revenue-adjusted rank                                 |
| M      | Raw PMC rank                                          |
| N      | Rank change after adjusting for revenue               |
| O      | Core-model inclusion flag: 1 = included; 0 = excluded |

The 2025 regression-control cells are:

| Cell | Field     |
|------|-----------|
| R32  | Slope     |
| R33  | Intercept |

| Cell | Field                        |
|------|------------------------------|
| R34  | Log-space R-squared          |
| R35  | Number of included companies |
| R36  | Optional model check         |

The existing model is:

$\text{LN}(1 + \text{PMC}) = \text{Intercept} + \text{Slope} \times \text{LN}(\text{Revenue})$

Current core formulas include:

$\text{D32} = \text{IF}(\text{C32} > 0, \text{LN}(\text{C32}), "")$

$\text{E32} = \text{IF}(\text{B32} < > "", \text{LN}(1 + \text{B32}), "")$

$\text{F32} = \text{IF}(\text{OR}(\text{D32} = "", \text{O32} < > 1), "", \$\text{R}\$33 + \$\text{R}\$32 * \text{D32})$

$\text{G32} = \text{IF}(\text{F32} = "", "", \text{EXP}(\text{F32}) - 1)$

$\text{H32} = \text{IF}(\text{F32} = "", "", \text{E32} - \text{F32})$

$\text{I32} = \text{IF}(\text{H32} = "", "", \text{EXP}(\text{H32}))$

$\text{J32} = \text{IF}(\text{I32} = "", "", \text{I32} - 1)$

$\text{K32} = \text{IF}(\text{G32} = "", "", \text{B32} - \text{G32})$

The regression formulas use only rows with positive revenue and an inclusion flag of 1.

Do not change the statistical method unless you identify a genuine error. Do not substitute simple PMC/revenue ratios for the regression.

## General instructions for every exhibit

Work one exhibit at a time.

For every exhibit, provide:

1. The analytical purpose and question answered.
2. The exact helper-table structure.
3. Exact Excel formulas using my current columns and rows where possible.
4. Instructions for copying or filling formulas.
5. Sorting and filtering instructions.

6. Exact Microsoft Excel chart type.
7. Step-by-step Excel menu instructions.
8. Axis assignments and axis titles.
9. Series names and recommended series formatting.
10. Data-label instructions, including "Value From Cells" when appropriate.
11. Recommended chart title.
12. Recommended footnote.
13. How to interpret the output.
14. Validation checks.
15. Common mistakes to avoid.

Use Microsoft Excel instructions, not Google Sheets instructions.

Do not provide vague formulas using names such as `ActualPMCCell`. Use explicit Excel references wherever the sheet layout is known. If a new summary sheet is needed, define its column layout first and then give exact formulas based on that layout.

Use blanks rather than zeros for excluded companies or missing financial data. Do not allow excluded companies to enter regression models or charts.

### **Exhibit 3: Actual versus revenue-expected PMC activity, 2025**

Build a clustered horizontal bar chart comparing:

- Actual PMC from column B
- Revenue-expected PMC from column G

Select approximately 10–12 companies using:

- The five largest positive actual-minus-expected differences from column K
- The five largest negative differences
- Important companies such as TMO or DHR if not already selected

Give exact instructions to create a helper table with:

- Ticker
- Actual PMC
- Expected PMC
- Actual-minus-expected difference
- Percentage above or below expected

Explain whether to sort by column K or another measure and why.

Use a clustered horizontal bar chart, not stacked bars.

Show how to:

- Reverse the ticker-axis order if necessary
- Add labels
- Distinguish actual from expected
- Use a normal linear axis initially
- Handle the chart if one very large company makes smaller bars unreadable
- Add a supporting table beside the chart

Use the title:

“Actual Versus Revenue-Expected PMC Activity, 2025”

## **Exhibit 4: Raw PMC methodology footprint, 2025**

Build a horizontal ranked bar chart using:

- Ticker from column A
- Actual PMC from column B
- Only companies with  $O = 1$

Sort from highest to lowest.

Explain why this chart remains necessary even after revenue adjustment: it measures absolute scientific footprint rather than intensity relative to size.

Provide instructions for displaying ticker labels, numerical values, and a clean axis.

Use the title:

“Raw PMC Methodology Footprint, 2025”

## **Exhibit 5: Raw rank versus revenue-adjusted rank, 2025**

Use:

- Adjusted rank from column L
- Raw rank from column M
- Rank change from column N

Create a horizontal rank-change bar chart.

Interpretation:

- Positive rank change = company moves up after adjusting for revenue
- Negative rank change = company moves down
- Zero = unchanged

Use the title:

“Change in PMC Ranking After Adjusting for Revenue, 2025”

Explain clearly that moving up after adjustment does not mean the company has a larger absolute footprint.

Also consider whether a dumbbell chart showing raw rank and adjusted rank would be more informative. Recommend one primary chart and explain why.

## Historical model construction: 2022–2025

After the 2025 exhibits, provide detailed instructions to duplicate the model separately for:

- 2022
- 2023
- 2024
- 2025

Each year must use:

- That year’s PMC count
- That year’s consolidated revenue
- Its own slope
- Its own intercept
- Its own R-squared
- Its own predicted PMC
- Its own residual
- Its own actual/expected multiple
- Its own adjusted ranking

Do not use 2025 coefficients for previous years.

Use the same core-business inclusion logic across years.

Explain how to handle:

- Companies missing revenue for one year
- Companies entering or leaving the public-company universe
- Acquisitions or divestitures
- Private companies
- Companies with zero PMC activity
- Companies that are included in some annual models but not all years

For longitudinal exhibits, recommend a primary balanced panel containing companies with valid data in all four years. Annual standalone models may contain all valid companies for the relevant year, but persistent four-year comparisons should prioritize a consistent universe.

Create a summary sheet named:

PMC 2022-2025 Summary

Include at minimum:

- Ticker
- 2022 actual PMC
- 2023 actual PMC
- 2024 actual PMC
- 2025 actual PMC
- 2022 residual
- 2023 residual
- 2024 residual
- 2025 residual
- 2022 activity multiple
- 2023 activity multiple
- 2024 activity multiple
- 2025 activity multiple
- Four-year cumulative PMC
- Average residual
- Persistent activity multiple
- Years above expectation
- Change in residual from 2022 to 2025
- Balanced-panel flag

Give exact formulas after defining the summary-sheet columns.

Do not average the four actual/expected multiples directly. Average the four log residuals and then calculate:

Persistent multiple =  $\text{EXP}(\text{Average residual})$

Explain why this is more statistically balanced.

## Exhibit 6: Raw PMC trend, 2022–2025

Build a line chart using actual PMC counts for approximately six to eight companies.

Include:

- Major scale leaders
- Important direct peers
- The primary company being analyzed
- One or two noteworthy smaller companies

Provide exact helper-table instructions.

Use the title:

“PMC Methodology Relationships, 2022–2025”

Explain why all companies should not be placed on one line chart.

Also propose a ranked 2022-to-2025 change chart as a supplemental visual.

## **Exhibit 7: Revenue-adjusted PMC trend, 2022–2025**

Build a line chart using annual actual/expected activity multiples.

- X-axis: 2022, 2023, 2024, 2025
- Y-axis: activity multiple
- Add a constant reference series of 1.0x
- Use approximately six to eight companies

Use the title:

“PMC Activity Relative to Revenue-Based Expectation, 2022–2025”

Explain how to interpret:

- Persistently above 1.0x
- Persistently below 1.0x
- Improving
- Deteriorating
- Volatile

Give exact instructions for formatting the y-axis as a multiple, such as 0.0x .

## **Exhibit 8: Four-year residual heatmap**

Use annual log residuals, not activity multiples.

Create a table with:

- Ticker
- 2022 residual
- 2023 residual
- 2024 residual
- 2025 residual

Use Excel conditional formatting with a three-color scale centered at zero.

Explain how to create a custom conditional-format rule where:

- Negative values indicate below expectation
- Zero is the neutral midpoint
- Positive values indicate above expectation

Use the title:

“Consistency of Revenue-Adjusted PMC Activity, 2022–2025”

Sort companies using either average residual or persistent multiple and explain the preferred method.

## **Exhibit 9: Persistent four-year adjusted ranking**

Calculate:

- Average log residual from 2022–2025
- Persistent activity multiple =  $\text{EXP}(\text{average residual})$
- Years above expectation
- Four-year cumulative raw PMC
- 2022-to-2025 residual change

Create a horizontal bar chart of the persistent multiple.

Use a reference line at 1.0x if practical.

Use the title:

“Persistent PMC Activity Relative to Company Size, 2022–2025”

Display or provide a supporting table containing:

- Ticker
- Persistent multiple
- Years above expectation
- Four-year raw PMC
- 2025 actual PMC
- 2025 activity multiple

Explain how to prevent a company with very low raw counts from appearing more important than a company with a large sustained scientific footprint.

## **Exhibit 10: PMC scale versus persistent adjusted performance**

Build a scatterplot using:

- X-axis: four-year cumulative PMC relationships

- Y-axis: persistent activity multiple
- Bubble size: 2025 consolidated revenue, if bubble charts work cleanly
- Data labels: ticker
- Horizontal reference line at 1.0x

Use the title:

"PMC Methodology Scale and Revenue-Adjusted Performance, 2022–2025"

Explain the four quadrants:

- High scale and above expectation
- High scale and below expectation
- Low scale and above expectation
- Low scale and below expectation

Explain whether the x-axis should be logarithmic and how to decide.

## Separate R&D-spend sensitivity analysis

After completing the revenue analysis, construct a separate R&D model. Do not combine revenue and R&D into one multivariable regression at this stage because the sample is small and revenue and R&D may be highly correlated.

Begin with a separate sheet named:

PMC R&D Model 2025

Use the same company universe where possible.

Replace revenue with reported R&D spending in millions of dollars.

Run:

$\text{LN}(1 + \text{PMC}) = \text{Intercept} + \text{Slope} \times \text{LN}(\text{R\&D spending})$

Give the full worksheet structure and exact formulas for:

- Actual PMC
- R&D spending
- LN R&D
- LN(1 + PMC)
- Slope
- Intercept
- R-squared
- Predicted LN PMC

- Expected PMC based on R&D
- Log residual
- Actual/expected multiple
- Percentage above or below expected
- Actual-minus-expected difference
- R&D-adjusted ranking
- Inclusion flag

Companies with missing or zero R&D must be excluded from the R&D regression and receive blank expected values.

Use reported consolidated R&D consistently unless directly comparable segment R&D is available for the complete universe.

Do not describe this as pure R&D productivity. Describe it as:

“PMC methodology activity relative to reported R&D spending.”

Explain the limitations:

- Current PMC use can reflect products developed years earlier
- Acquired products may not reflect the current company's historical R&D
- Current R&D may fund products that have not yet generated publications
- Consolidated R&D can include unrelated businesses

## **R&D Exhibit 1: PMC versus R&D spending, 2025**

Build a log-log scatterplot comparable to the revenue exhibit.

Use:

- X-axis: 2025 R&D spending
- Y-axis: Actual PMC + 1
- Fitted expected series from the controlled worksheet regression
- Ticker labels
- Logarithmic axes

Use the title:

“PMC Methodology Activity Relative to R&D Spending, 2025”

Report the worksheet-calculated log-space R-squared rather than relying on Excel's chart trendline R-squared.

## **R&D Exhibit 2: R&D-adjusted ranking**

Create a horizontal bar chart of percentage above or below R&D-based expectation.

Show actual PMC and R&D-expected PMC in a supporting table.

Use the title:

“PMC Activity Above/(Below) R&D-Based Expectation, 2025”

## **Revenue-versus-R&D comparison exhibit**

Create a scatterplot with:

- X-axis: revenue-model log residual
- Y-axis: R&D-model log residual
- Ticker labels
- Vertical reference line at zero
- Horizontal reference line at zero

Use residuals, not percentages, because zero is a symmetrical neutral point.

Use the title:

“PMC Activity Relative to Revenue and R&D Benchmarks, 2025”

Explain the four quadrants:

- Positive on both: above expectation versus both company size and R&D
- Positive revenue residual, negative R&D residual
- Negative revenue residual, positive R&D residual
- Negative on both

Give exact Excel instructions for adding the zero-reference lines as additional series.

## **Historical R&D analysis**

If R&D data are complete and consistently defined for 2022–2025, replicate the separate annual R&D regressions.

Create:

- Annual R&D-adjusted activity multiples
- R&D residual heatmap
- Persistent R&D-adjusted multiple
- Revenue-persistent versus R&D-persistent comparison

If R&D disclosure is inconsistent across years, keep the R&D analysis as a 2025 sensitivity test and clearly disclose that limitation.

## Optional PMC breadth analysis after the core exhibits

Only after the primary scale and normalization exhibits are complete, outline possible PMC-only breadth analyses:

- Unique journals by company and year
- Unique canonical product families
- PMC relationships per journal
- PMC relationships per product family
- Top-product concentration
- Top-three-product concentration
- Top-journal concentration
- Top-five-journal concentration

Do not use raw matched aliases as product breadth without canonicalizing product names.

These breadth exhibits should remain within the PMC analysis and should not incorporate other data sources.

## Methodological rules

Preserve these rules throughout:

- One cleaned company-publication relationship is the unit of PMC activity.
- Do not count duplicate mentions within the same company-publication relationship.
- Do not use simple PMC/revenue as the primary adjusted metric.
- Use the log-log regression and log residual as the main normalization framework.
- Use raw PMC counts alongside adjusted metrics so scale is never lost.
- Exclude companies based on business-model incompatibility or unavailable data, not merely because their PMC count is low.
- Maintain a written reason for every exclusion.
- Keep the included-company universe as consistent as possible across years.
- Do not describe above-expectation PMC activity as proof of better innovation, product quality, or R&D productivity.
- Use language such as “observable PMC methodology presence,” “activity relative to expectation,” and “scientific-methodology footprint.”
- Report the number of included companies and log-space R-squared for each annual model.
- Use the spreadsheet-calculated model statistics rather than Excel’s chart-generated R-squared when they differ.
- Check that a displayed fitted equation’s exponent matches the regression slope and its constant matches EXP(intercept).

Begin with Exhibit 3 only. Give me its complete formulas and Microsoft Excel instructions, then stop and wait for me to confirm before proceeding to Exhibit 4.